L D COLLEGE OF ENGINEERING

Computer Engineering Department

Implementation of a Fuzzy Controller in Python For Smart Traffic Signal Controller

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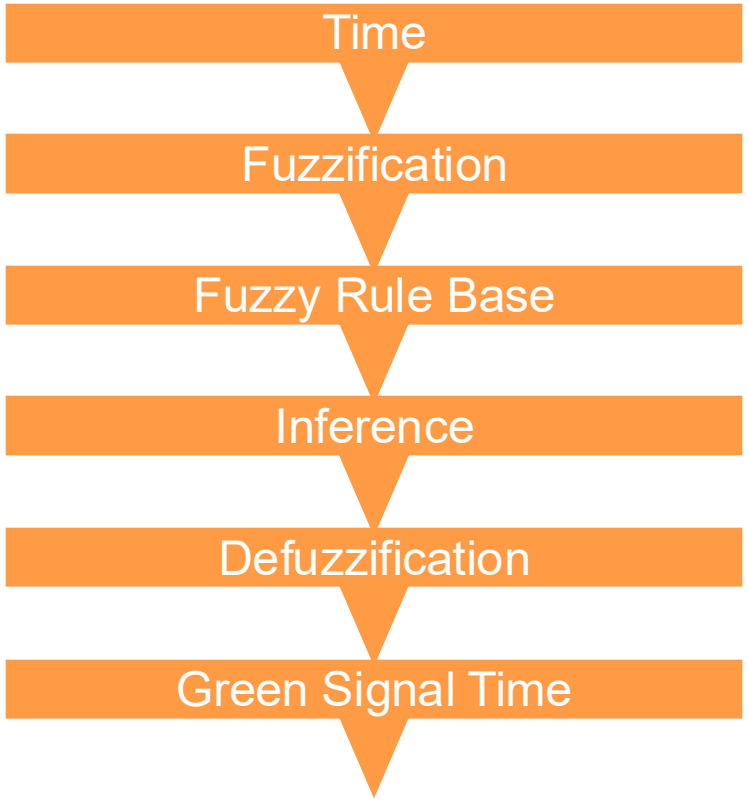
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Aim of Implementation

- Implement a fuzzy controller using Python.
- Use scikit-fuzzy for fuzzy logic.
- Automatically determine suitable green signal time.
- Use traffic density and waiting time as inputs.

System Architecture Traffic Density + Waiting Time



Python Libraries

- Python

```
import numpy as np
import skfuzzy as fuzz
from skfuzzy import control as ctrl
```

- NumPy → numerical operations
- scikit-fuzzy → fuzzy logic
- control → fuzzy control system

Define Input & Output Variables

- Python

```
Density = ctrl.Antecedent(
    np.arange(0, 101, 1), ‘density’)

waiting = ctrl.Antecedent(
    np.arange(0, 121, 1), ‘waiting’)

green = ctrl.Consequent(
    np.arange(10, 91, 1), ‘green’)
```

- *Density: 0–100%*
- *Waiting time: 0–120 sec*
- *Green time: 10–90 sec*

Traffic Density Membership Functions

- Low
- Medium
- High
- Triangular membership function

```
Python

Density[‘low’] = fuzz.trimf(
    density.universe, [0, 0, 50])

density[‘medium’] = fuzz.trimf(
    density.universe, [25, 50, 75])

density[‘high’] = fuzz.trimf(
    density.universe, [50, 100, 100])
```

Waiting Time Membership Functions

- Short
- Medium
- Long
- Range: 0–120 seconds

- Python

```
Waiting['short'] = fuzz.trimf(
    waiting.universe, [0, 0, 60])

waiting['medium'] = fuzz.trimf(
    waiting.universe, [30, 60, 90])

waiting['long'] = fuzz.trimf(
    waiting.universe, [60, 120, 120])
```

Green Signal Membership Functions Five output sets:

- Very Short → 10 sec
- Short → 30 sec
- Medium → 50 sec
- Long → 70 sec
- Very Long → 90 sec

Fuzzy Rules Example:

- Python

```
rule1 = ctrl.Rule(
    density['low'] & waiting['short'],
    green['very_short'])



rule9 = ctrl.Rule(
    density['high'] & waiting['long'],
    green['very_long'])
```

Create & Run Controller

- Python

```
traffic_ctrl = ctrl.ControlSystem(rules)

traffic_sim =
ctrl.ControlSystemSimulation(
traffic_ctrl)

traffic_sim.input['density'] = 65
traffic_sim.input['waiting'] = 75
traffic_sim.compute()
```

Output & Result

- Python

```
Print(traffic_sim.output['green'])
```

Sample input:

Density = 65%

Waiting = 75sec

Final green signal time:

≈ 68 seconds

Conclusion

- Fuzzy controller successfully implemented in Python.
- Scikit-fuzzy provides the fuzzy-control functionality.
- Green time changes according to traffic conditions.
- The controller provides an adaptive alternative to fixed-time signals.
- Final test: 65% density + 75 sec waiting → ≈68 sec green time.

THANK YOU